PENG - Penguin Solutions

Full investment research | Model assumptions and source evidence

Framework 8.0 | Research cutoff: 7 October 2026 | Fiscal Q4 and FY2026 update

Q4 2027 operating ceiling: USD 118.18 per share. Previous same-quarter ceiling: USD 97.29. The valuation remains **20x forward-four-quarter GAAP operating profit**, using the labeled quarter plus the next three quarters and the labeled-quarter economic-share denominator.

Portfolio Excel Base: 73/100, unchanged. Analytical Base: 72/100 (43/60), unchanged. Financial amounts are USD millions, shares are millions and ceilings are USD per share unless stated otherwise.

Status: Full core research artifact updated for October 6 earnings, FY27 guidance, financing cash flows and legacy capped-call evidence. The realized August quarter is a reported anchor; later operating rows are probability-weighted assumptions. Portfolio Base is a separate protected judgment, not a probability multiplier.

The Q4 2027 ceiling rises from **$97.29 to $118.18**. Updating operating economics alone gives **$116.30** with the previous share denominator. A 90% probability-weighted legacy hedge credit adds the remaining **$1.88**. The multiple is unchanged. Faster memory and AI-integration growth is accompanied by substantial inventory funding and negative operating cash flow.

1. Thesis, products and competitive position

AI INFRASTRUCTURE AND MEMORY SOLUTIONS. Penguin combines memory modules, AI-system integration and operating services. This creates opportunities in enterprise inference, but does not make all revenue recurring software or give it the economics of a DRAM manufacturer.

Fiscal Q4 FY2026, ended August 28, reported revenue of $566.685m, GAAP gross profit of $155.900m, opex of $86.438m and operating profit of $69.462m. Adjusted OP of $89.796m is a separate measure. The valuation retains GAAP operating profit including recurring share compensation. Equity-investment gains, financing proceeds and income-tax effects are excluded from OP. Sources: PENG-8 and PENG-9.

Distinct sources of revenue and margin

Penguin combines enterprise/AI infrastructure integration, managed operations, SMART modular memory-related activity and other products. ClusterWare and deployment expertise can support repeat service relationships, while hardware procurement and memory logistics generally capture a different margin from software. The five-vector model keeps those economics separate.

MemoryAI and CXL-related approaches address inference memory capacity and movement, including KV-cache use cases. The original performance, latency and cost claims are vendor/workload-specific demonstrations, not universal benchmarks across all AI deployments. Commercial contribution requires qualified products, customer usage and measurable paid content.

Procurement is not all revenue

Gross memory throughput through a logistics program can greatly exceed recognized net fees. The model must follow the principal-versus-agent treatment rather than count the full purchased memory value and then also count service fees. Procurement terms, replacement costs, inventory funding and customer collections are critical to cash conversion during a fast memory-price cycle.

The original Celestial/Marvell and other ecosystem relationships help explain future memory and interconnect possibilities; they are not unconditional Penguin revenue commitments. Hardware integration is distinct from owning and financing a large GPU-cloud asset base. GAAP and adjusted operating-profit measures also need an explicit reconciliation; the current anchor below identifies the selected basis.

Research provenance: the product and competitive analysis in this section synthesizes the supplied original research. Dated market-share, process and channel claims retain their original evidence status; this integration is not a new independent verification of those claims. Current earnings and model assumptions are set out below.

2. Earnings, guidance and operating assumptions

Latest results and guidance

The October 6 release covers fiscal Q4 and FY2026 ended August 28. Q4 segment revenue was $340.784m Memory, $154.039m Advanced Computing and $71.862m LED; full-year revenue was $924.001m, $558.789m and $248.678m, respectively. These are reported segments. The five-product split later in this document is an analytical allocation.

Metric (USD m except EPS/shares)	Q4 FY26 reported	FY26 reported
Revenue	566.685	1,731.468
GAAP gross profit	155.900	478.925
GAAP operating expenses	86.438	313.329
GAAP operating profit	69.462	165.596
GAAP / adjusted gross margin	27.5% / 28.8%	27.7% / 29.3%
Adjusted operating profit	89.796	240.962
GAAP / adjusted EPS	1.29 / 1.00	2.60 / 2.87
GAAP / adjusted average diluted shares	64.040 / 62.000	58.825 / 55.977
Continuing operating cash flow	(163.136)	(151.914)
Capex including equipment deposits	4.272	11.569

July's FY26 guide was revenue growth of 22% +/-2 percentage points, GAAP gross margin of 26.5% +/-0.5 and GAAP opex of $303m +/-$5m. Actual growth was approximately 26%, margin 27.7% and opex $313.329m. Revenue and profit improved, but costs also exceeded the prior range. Adjusted EPS of $2.87 exceeded the old $2.60 +/-$0.05 guide. Sources: PENG-1, PENG-8 and PENG-10.

October provides a fuller FY27 guide than July's initial planning view. Management guides Advanced Computing growth to 40% +/-10 percentage points, Memory to 50% +/-10 points and LED approximately flat. There is no issuer quarterly revenue cadence, unit shipment path or full cash-flow forecast.

FY27 issuer guidance	GAAP	Adjusted
Revenue growth from FY26 actual	40% +/-10 percentage points	Same revenue basis
Gross margin	27% +/-2 percentage points	28% +/-2 percentage points
Operating expenses	329 +/-10	275 +/-10
Diluted EPS	3.50 +/-0.70	4.45 +/-0.70
Weighted-average diluted shares	63m	63m
Tax rate	Not separately guided here	20%

Management calls the revenue midpoint about $2.43bn. Exact arithmetic from $1,731.468m x1.40 gives **$2,424.0552m**, used in this model. The previous $2.17bn midpoint used 30% growth from the old FY26 guidance, so the percentage comparisons have different denominators. Rounded segment guidance sums slightly below the exact consolidated midpoint; the conditional segment mix is scaled by 1.0029256 to reconcile it rather than invent another revenue source.

FY27 Outlook

	GAAP Outlook	Adjustments	Non-GAAP Outlook
Net sales	40% YoY Growth +/- 10%	—	40% YoY Growth +/- 10%
Gross margin	27% +/- 2%	1% (A)	28% +/- 2%
Operating expenses	$329 million +/- $10 million	$(54) million (B)(C)	$275 million +/- $10 million
Diluted earnings per share	$3.50 +/- $0.70	$0.95 (A)(B)(C)(D)(E)	$4.45 +/- $0.70
Diluted shares	63 million	—	63 million

Non-GAAP adjustments (in millions)	
(A) Stock-based compensation and amortization of acquisition-related intangibles included in cost of sales	$ 30
(B) Stock-based compensation and amortization of acquisition-related intangibles included in R&D and SG&A	46
(C) Other operating adjustments	8
(D) Estimated income tax effects	(17)
(E) Estimated effect of allocation of earnings to participating securities	(7)
	$ 60

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Issuer evidence: original October 6 presentation slide 27, FY27 guidance. Financial columns are management guidance, not ASTRA forecasts. Source PENG-9.

Customer evidence, cash conversion and source limitations

Management reported six new AI customers in Q4, including four neoclouds; 17 new AI customers and 12 expansions in FY26; and 15 new Memory customers plus 29 expansions for the year. These support the guidance branch but receive **100% overlap** with it, not a second revenue overlay. The cited $3bn and $10bn contract values belong to customers, not Penguin. A 36,000-GPU Norway deployment describes project scale, not disclosed Penguin sales.

Prepared remarks say non-hyperscale AI was 66% of Q4 Advanced Computing and grew 99% year over year. They also say prior-year mix was 60%, which is incompatible with those figures and reported segment totals. The first two imply $101.67m currently and $51.09m previously, or 36.9% of the prior $138.336m segment. Slide 13 repeats 66% and 99%, but not 60%. The model does not construct a historical series from the inconsistent prior-year claim.

Inventory reached $748.785m and receivables $796.264m. Continuing operating cash flow was negative $163.136m for Q4 and $151.914m for FY26. Year-end cash was $647.208m, but growth still requires substantial funding. Working-capital metrics use approximately $1.574bn gross sales and $1.418bn gross COGS because memory logistics is reported net as agent. Applying DSO/DIO/DPO to $566.685m net revenue would understate the funding cycle. Sources: PENG-8 to PENG-10.

Q4 GAAP EPS includes a $57.595m tax benefit. Adjusted reconciliation removes $75.907m of tax effects and adds back $33.248m of conversion inducement expense. Neither becomes a recurring product-margin assumption. The current basis remains GAAP operating profit; adjusted results are shown separately.

Rule 16b: bounded company economics

The previous model is the reference. The FY27 conditional case uses the issuer's exact consolidated revenue midpoint, segment mix, 27% GAAP gross margin and $329m opex. The incremental case has **75% probability**, with **25% retaining the reference**. Specific guidance and customer additions support realization; availability, acceptance, customer funding and negative operating cash flow prevent certainty. These are analyst judgments, not source-authored probabilities.

The reference seasonal allocation is retained. Conditional revenue factors are 1.0842873 for core memory/net logistics, 1.2119143 for each of three Advanced Computing subcategories and 1.0496866 for LED. Product margins are scaled by 0.9863314 to reconcile FY27 consolidated gross margin. Fixed opex is scaled by 0.8599093 after retaining 4.5% of revenue as variable expense; conditional annual opex equals $329m. Reference and conditional product gross profits and expenses are weighted jointly, avoiding unrelated weighted-margin arithmetic or double-counted operating leverage.

FY27: Q4 2026-Q3 2027	Reference	Conditional midpoint	Active weighted
Revenue	2,167.00	2,424.06	2,359.79
Gross profit	590.87	654.49	638.59
Operating expenses	353.26	329.00	335.07
GAAP OP	237.61	325.49	303.52

Later periods use the same relative product/cost mechanism as a persistence scenario, not new guidance. Joint incremental probabilities decline to 60% in FY28, 45% in FY29, 30% in FY30 and 25% in the modeled first quarter of FY31. They apply directly to conditional-minus-reference economics, without multiplication by 75% again. The reference already has substantial MemoryAI/service growth; declining persistence reduces the additional October upgrade, not the pre-existing adoption assumptions.

Period / calendar window	Joint probability	Conditional bound and counter-case
FY27: Q4 2026-Q3 2027	75%	Issuer midpoint; 25% reference
FY28: Q4 2027-Q3 2028	60%	Same relative uplift; 40% reference
FY29: Q4 2028-Q3 2029	45%	Same relative uplift; 55% reference

Period / calendar window	Joint probability	Conditional bound and counter-case
FY30: Q4 2029-Q3 2030	30%	Same relative uplift; 70% reference
FY31 partial: Q4 2030	25%	Same relative uplift; 75% reference

Issuer revenue guidance spans $2,250.9084m-$2,597.2020m, gross margin 25%-29%, and opex $319m-$339m. Combining all low or all high endpoints mechanically gives $223.73m-$434.19m operating profit. That is an outer sensitivity envelope, not an issuer-guided OP range or independent outcomes: inputs can be correlated. The active model uses the defined midpoint-versus-reference branch.

Existing product and capacity assumptions

Calendar Q3 maps to fiscal August, Q4 to November, Q1 to February and Q2 to May. FY27 guidance covers Q4 2026-Q3 2027; the active Q4 2027 ceiling covers Q4 2027-Q3 2028. Calendar annual totals cannot be compared directly with fiscal guidance.

The five vectors remain core memory/net logistics, hardware, MemoryAI, software/services and LED. New products are carved out of parent segments, never added on top of them. Prior calendar 2027-30 reference revenue was $1,300m/$1,550m/$1,700m/$1,800m for memory, $500m/$650m/$820m/$1,050m for hardware, $60m/$180m/$350m/$550m for MemoryAI, $100m/$170m/$240m/$330m for software/services and $240m/$220m/$210m/$200m for LED. These are retained reference figures, not active post-update forecasts.

Reference 2028 memory growth assumed approximately 15% more shipped content and 3.7% ASP/mix. Hardware growth assumed 20% deployment volume and 8.3% configuration value. MemoryAI's $180m reference requires three times its 2027 base; subsequent 94% and 57% growth require paid repeat deployments. Software/services growth requires attachments and monetization; not all group services are recurring software. Absolute rack, appliance and memory-bit capacity is insufficiently disclosed to certify these paths. No separate scarcity premium is added on top of the guidance without distinct evidence.

Reference 2027-30 margins were 26/26/27/27% for memory, 24/24/25/26% for hardware, 40/42/44/45% for MemoryAI, 55/58/60/62% for software/services and 28% for LED. Original opex used $60m quarterly fixed cost growing 10% annually plus 4.5% of revenue. Active weighted margins and opex follow the calibration above and the tables below.

Financing and economic-share reconciliation

July issued $750m of 0% notes due 2031. October cash-flow reporting records issuance and $295.454m repayment of old notes; remarks confirm completion of about $296m of exchanges. July's 8-K described approximately 8.7m exchange shares. Exact final common shares remain a filing checkpoint. Financing proceeds fund obligations and operations; they are not revenue or OP.

The established share proxy starts with July 2 common of 51.242699m, adds 8.7m exchange shares, 6.095703m preferred-conversion equivalents at $32.81 and a 2m outstanding-award allowance, then 1m annual net awards from 2027. Preferred conversion is a successful-case assumption, not a reported event. The 63m FY27 weighted-average guide and Q4 64.040m GAAP/62.000m adjusted shares use different accounting and period bases, and do not replace the economic denominator.

The reference included residual $14.5m 2029 and $40m 2030 note principal and excess-share dilution, but no old hedge offset. July 13 issuer disclosure says it did not intend to terminate or amend existing capped calls, while preserving future unwind discretion. October slide 32 reconciles to residual old-note dilution and approximately $125.1m full legacy capped-call economics divided by reference price. At $100, it shows 1.568m gross dilution, 1.251m call offset and 0.317m net; at $125 it shows 1.677m, 1.001m and 0.676m. Source quantities are rounded; final settlement remains a gap.

The active model credits **90% of min(positive gross legacy dilution, $125.1m/reference price)** from 2027, with 10% no credit for survival, settlement and counterparty uncertainty. It does not turn negative hedge economics into negative issued shares and does not also add a cash asset for the same hedge. At the 2027 $100 reference price, credit is 1.1259m, lowering the denominator from 70.604573m to 69.478673m. The 2026 denominator is unchanged.

The separate new-note curve stays 6.42675m x max(0, 1-$175.05/reference price). Annual reference prices remain $100/$125/$175/$225 for 2027-30; these are dilution inputs, not market prices or targets. The legacy slide stops at $175 and is not the entire capital structure above that value. Principal cash settlement, hedge realization and awards can change economic shares. The existing OP valuation convention is retained; this update introduces no separate net-cash asset uplift or change in claim accounting.

Convertible Dilution

	(In Thousands of Shares)		
Stock Price	Share Dilution	Capped Calls	Dilution to PENG
$16	—	—	—
$17	—	—	—
$18	—	—	—
$19	—	—	—
$20	—	—	—
$21	—	—	—
$22	24	(248)	(224)
$23	53	(544)	(491)
$24	79	(816)	(737)
$25	103	(1,066)	(963)
$26	126	(1,297)	(1,171)
$27	146	(1,510)	(1,364)
$28	166	(1,709)	(1,543)
$29	232	(2,138)	(1,906)
$30	295	(2,337)	(2,042)
$31	354	(2,492)	(2,138)
$32	409	(2,637)	(2,228)
$33	460	(2,773)	(2,313)
$34	509	(2,902)	(2,393)
$35	555	(3,023)	(2,468)
$36	598	(3,137)	(2,539)
$37	639	(3,246)	(2,607)
$38	678	(3,292)	(2,614)
$39	715	(3,208)	(2,493)
$40	750	(3,128)	(2,378)
$41	783	(3,052)	(2,269)
$42	815	(2,979)	(2,164)
$43	845	(2,910)	(2,065)
$44	874	(2,843)	(1,969)
$45	901	(2,780)	(1,879)
$46	928	(2,720)	(1,792)
$47	953	(2,662)	(1,709)
$48	977	(2,607)	(1,630)
$49	1,000	(2,553)	(1,553)
$50	1,022	(2,502)	(1,480)
$51	1,044	(2,453)	(1,409)
$52	1,064	(2,406)	(1,342)
$53	1,084	(2,361)	(1,277)
$54	1,103	(2,317)	(1,214)
$55	1,122	(2,275)	(1,153)

	(In Thousands of Shares)		
Stock Price	Share Dilution	Capped Calls	Dilution to PENG
$56	1,139	(2,234)	(1,095)
$57	1,156	(2,195)	(1,039)
$58	1,173	(2,157)	(984)
$59	1,189	(2,121)	(932)
$60	1,204	(2,085)	(881)
$61	1,219	(2,051)	(832)
$62	1,234	(2,018)	(784)
$63	1,248	(1,986)	(738)
$64	1,261	(1,955)	(694)
$65	1,274	(1,925)	(651)
$66	1,287	(1,896)	(609)
$67	1,299	(1,867)	(568)
$68	1,311	(1,840)	(529)
$69	1,323	(1,813)	(490)
$70	1,334	(1,787)	(453)
$71	1,345	(1,762)	(417)
$72	1,356	(1,738)	(382)
$73	1,366	(1,714)	(348)
$74	1,376	(1,691)	(315)
$75	1,386	(1,668)	(282)
$76	1,396	(1,646)	(250)
$77	1,405	(1,625)	(220)
$78	1,414	(1,604)	(190)
$79	1,423	(1,584)	(161)
$80	1,432	(1,564)	(132)
$81	1,440	(1,545)	(105)
$82	1,448	(1,526)	(78)
$83	1,456	(1,507)	(51)
$84	1,464	(1,489)	(25)
$85	1,472	(1,472)	—
$86	1,479	(1,455)	24
$87	1,486	(1,438)	48
$88	1,493	(1,422)	71
$89	1,500	(1,406)	94
$90	1,507	(1,390)	117
$91	1,514	(1,375)	139
$92	1,520	(1,360)	160
$93	1,527	(1,345)	182
$94	1,533	(1,331)	202
$95	1,539	(1,317)	222

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Issuer evidence: original slide 32, left detail panel, prices $16-$95. Negative net values describe hedge economics, not negative issued shares. Source PENG-9; every numerical column is preserved across both detail panels.

	(In Thousands of Shares)		
Stock Price	Share Dilution	Capped Calls	Dilution to PENG
$96	1,545	(1,303)	242
$97	1,551	(1,290)	261
$98	1,557	(1,277)	280
$99	1,562	(1,264)	298
$100	1,568	(1,251)	317
$101	1,573	(1,239)	334
$102	1,579	(1,227)	352
$103	1,584	(1,215)	369
$104	1,589	(1,203)	386
$105	1,594	(1,192)	402
$106	1,599	(1,180)	419
$107	1,604	(1,169)	435
$108	1,608	(1,158)	450
$109	1,613	(1,148)	465
$110	1,617	(1,137)	480
$111	1,622	(1,127)	495
$112	1,626	(1,117)	509
$113	1,631	(1,107)	524
$114	1,635	(1,097)	538
$115	1,639	(1,088)	551
$116	1,643	(1,079)	564
$117	1,647	(1,069)	578
$118	1,651	(1,060)	591
$119	1,655	(1,051)	604
$120	1,659	(1,043)	616
$121	1,663	(1,034)	629
$122	1,666	(1,026)	640
$123	1,670	(1,017)	653
$124	1,673	(1,009)	664
$125	1,677	(1,001)	676
$126	1,680	(993)	687
$127	1,684	(985)	699
$128	1,687	(977)	710
$129	1,691	(970)	721
$130	1,694	(962)	732
$131	1,697	(955)	742
$132	1,700	(948)	752
$133	1,703	(941)	762
$134	1,706	(934)	772
$135	1,709	(927)	782

	(In Thousands of Shares)		
Stock Price	Share Dilution	Capped Calls	Dilution to PENG
$136	1,712	(920)	792
$137	1,715	(913)	802
$138	1,718	(907)	811
$139	1,721	(900)	821
$140	1,724	(894)	830
$141	1,726	(887)	839
$142	1,729	(881)	848
$143	1,732	(875)	857
$144	1,735	(869)	866
$145	1,737	(863)	874
$146	1,740	(857)	883
$147	1,742	(851)	891
$148	1,745	(845)	900
$149	1,747	(840)	907
$150	1,750	(834)	916
$151	1,752	(829)	923
$152	1,754	(823)	931
$153	1,757	(818)	939
$154	1,759	(812)	947
$155	1,761	(807)	954
$156	1,764	(802)	962
$157	1,766	(797)	969
$158	1,768	(792)	976
$159	1,770	(787)	983
$160	1,772	(782)	990
$161	1,775	(777)	998
$162	1,777	(772)	1,005
$163	1,779	(768)	1,011
$164	1,781	(763)	1,018
$165	1,783	(758)	1,025
$166	1,785	(754)	1,031
$167	1,787	(749)	1,038
$168	1,789	(745)	1,044
$169	1,791	(740)	1,051
$170	1,792	(736)	1,056
$171	1,794	(732)	1,062
$172	1,796	(727)	1,069
$173	1,798	(723)	1,075
$174	1,800	(719)	1,081
$175	1,802	(715)	1,087

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Issuer evidence: original slide 32, right detail panel, prices $96-$175. Legacy note/call economics are separate from the 2031 note and $175.05 cap branch. Source PENG-9.

Year-end debt carrying amounts were $53.418m current and $735.532m long term; preferred temporary equity was $202.710m. Cash was $647.208m, or $647.524m including restricted cash. Carrying debt is not face principal; preferred carrying value is not automatically conversion value; restricted cash is not freely available. Negative cash flow and inventory/receivables remain material risks.

CFO compensation includes $550k salary, 90% target bonus, $700k retention in two installments and $4.3m RSU grant value. The model assigns 100% overlap with issuer opex guidance and the existing award proxy. It adds no duplicate expense or unpriced grant-value-to-share conversion. Offer letter, vesting and exact granted shares remain checkpoints.

3. Analytical score

The category arithmetic gives **43/60 = 72/100** after the existing -5-point risk deduction. The portfolio Base remains **73/100**. Risk-class detail appears in section 7.

Category	Points	Max	Basis
Core thesis	16	20	AI integration, memory procurement and services expertise; limited platform control.
Cycle stage	8	10	Memory demand and customer diversification improve, with pricing-driven gains.
TAM expansion	8	10	CXL and inference memory extend the opportunity but remain small products.
Catalysts	5	5	New logos, MemoryAI deployments and additional financing capacity.
Macro	8	10	AI deployment supports demand; component shortages also constrain delivery.
Alpha	3	5	Net logistics revenue, physical modules and overlapping platform categories reconciled.
Risks	-5	0	Hardware margin, memory cycle, working capital, dilution and execution.

4. Product economics and quarterly operating model

Q3 2026 consolidated revenue, GP, opex and OP are reported. Memory and LED revenue match reported segments. The three Advanced Computing subcategories and all product GP/margins are analytical allocations reconciled to actual totals. Later rows are probability-weighted model assumptions, not issuer quarterly guidance.

Vector evidence

Vector	Source / derivation	Confidence
Memory / net logistics	Q4 segment $340.784m; agent logistics net	High actual; low ASP/volume split
Computing hardware	Q4 AC total $154.039m includes services	High parent segment; low submix
MemoryAI systems	CXL/KV-cache wins, no separate sales run rate	Low dollar forecast
Software / services	Q4 group services about $57m, not all software	High broad total; low premium mix
Optimized LED	Q4 revenue $71.862m	High actual; low future margins

Product revenue build (USD millions)

Quarter	Memory / logistics	Hardware	MemoryAI	Software / services	LED	Total
Q3 2026	340.78	100.92	10.62	42.49	71.86	566.68
Q4 2026	340.23	121.69	13.91	48.68	62.24	586.74
Q1 2027	317.90	133.28	15.99	26.66	57.26	551.08
Q2 2027	331.72	139.07	16.69	27.81	59.75	575.05
Q3 2027	373.19	156.46	18.77	31.29	67.21	646.93
Q4 2027	355.09	146.53	17.58	29.31	64.26	612.77
Q1 2028	374.53	168.51	46.66	44.07	52.11	685.88
Q2 2028	390.81	175.84	48.69	45.99	54.37	715.70
Q3 2028	439.66	197.81	54.78	51.74	61.17	805.17
Q4 2028	418.29	185.12	51.26	48.41	58.48	761.56
Q1 2029	405.83	206.59	88.18	60.46	49.38	810.44
Q2 2029	423.48	215.57	92.01	63.09	51.53	845.67
Q3 2029	476.41	242.51	103.51	70.98	57.97	951.38
Q4 2029	453.18	226.75	96.79	66.37	55.41	898.50
Q1 2030	424.47	256.85	134.54	80.73	46.69	943.27
Q2 2030	442.92	268.02	140.39	84.24	48.72	984.29
Q3 2030	498.29	301.52	157.94	94.76	54.80	1,107.32
Q4 2030	477.86	287.46	150.58	90.35	52.65	1,058.89

Product gross margins (analytical allocations)

Quarter	Memory / logistics	Hardware	MemoryAI	Software / services	LED
Q3 2026	25.7%	21.0%	46.8%	60.8%	22.6%
Q4 2026	23.7%	21.8%	39.6%	52.4%	27.7%
Q1 2027	25.7%	23.7%	39.6%	54.4%	27.7%
Q2 2027	25.7%	23.7%	39.6%	54.4%	27.7%
Q3 2027	25.7%	23.7%	39.6%	54.4%	27.7%
Q4 2027	25.8%	23.8%	39.6%	54.5%	27.8%
Q1 2028	25.8%	23.8%	41.6%	57.5%	27.8%
Q2 2028	25.8%	23.8%	41.6%	57.5%	27.8%
Q3 2028	25.8%	23.8%	41.6%	57.5%	27.8%
Q4 2028	25.8%	23.8%	41.7%	57.6%	27.8%
Q1 2029	26.8%	24.8%	43.7%	59.6%	27.8%
Q2 2029	26.8%	24.8%	43.7%	59.6%	27.8%
Q3 2029	26.8%	24.8%	43.7%	59.6%	27.8%
Q4 2029	26.9%	24.9%	43.8%	59.7%	27.9%
Q1 2030	26.9%	25.9%	44.8%	61.7%	27.9%
Q2 2030	26.9%	25.9%	44.8%	61.7%	27.9%
Q3 2030	26.9%	25.9%	44.8%	61.7%	27.9%
Q4 2030	26.9%	25.9%	44.8%	61.8%	27.9%

Quarterly operating model

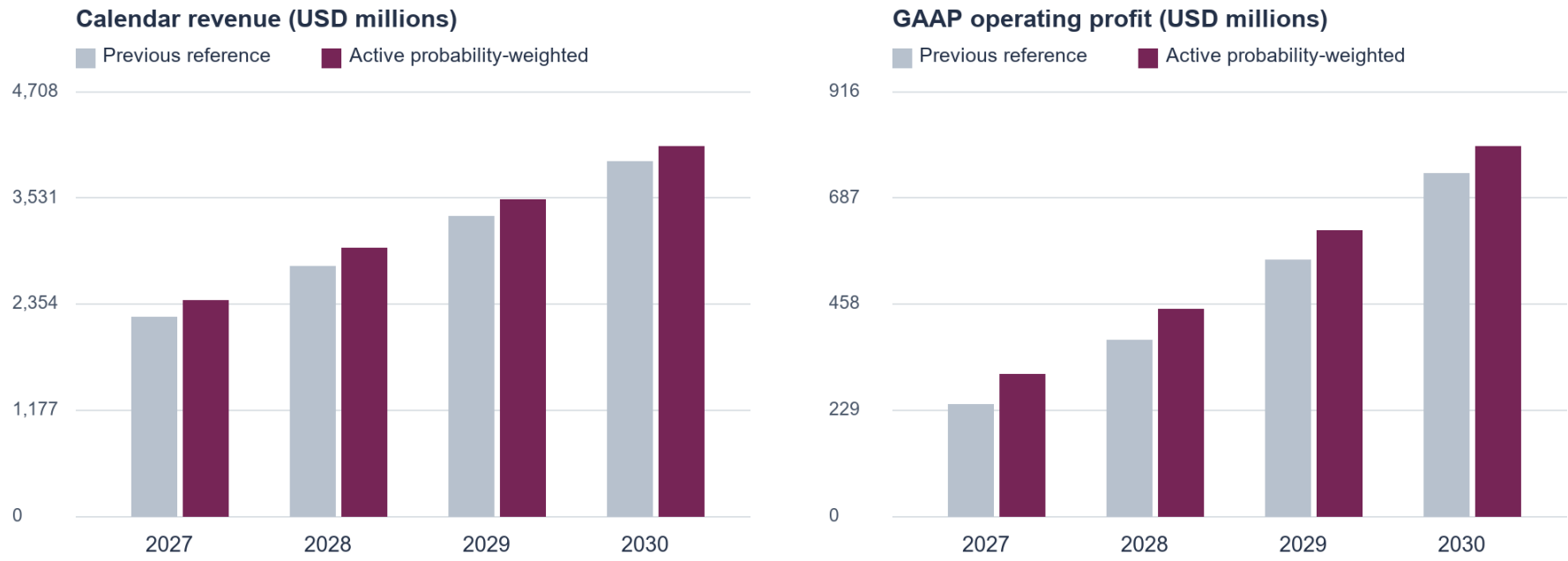
Quarter	Status	Revenue	GP	GM	Opex	GAAP OP	OP margin
Q3 2026	Actual total	566.68	155.90	27.5%	86.44	69.46	12.3%
Q4 2026	Model, 75% branch	586.74	155.56	26.5%	78.08	77.47	13.2%
Q1 2027	Model, 75% branch	551.08	150.13	27.2%	83.86	66.27	12.0%
Q2 2027	Model, 75% branch	575.05	156.66	27.2%	84.94	71.72	12.5%
Q3 2027	Model, 75% branch	646.93	176.24	27.2%	88.18	88.06	13.6%
Q4 2027	Model, 60% branch	612.77	167.19	27.3%	88.03	79.16	12.9%
Q1 2028	Model, 60% branch	685.88	195.87	28.6%	97.36	98.51	14.4%
Q2 2028	Model, 60% branch	715.70	204.39	28.6%	98.70	105.68	14.8%
Q3 2028	Model, 60% branch	805.17	229.93	28.6%	102.73	127.20	15.8%
Q4 2028	Model, 45% branch	761.56	217.73	28.6%	102.29	115.43	15.2%
Q1 2029	Model, 45% branch	810.44	248.47	30.7%	111.30	137.17	16.9%
Q2 2029	Model, 45% branch	845.67	259.27	30.7%	112.88	146.39	17.3%
Q3 2029	Model, 45% branch	951.38	291.68	30.7%	117.64	174.04	18.3%
Q4 2029	Model, 30% branch	898.50	275.72	30.7%	116.94	158.79	17.7%
Q1 2030	Model, 30% branch	943.27	303.67	32.2%	126.60	177.07	18.8%

Quarter	Status	Revenue	GP	GM	Opex	GAAP OP	OP margin
Q2 2030	Model, 30% branch	984.29	316.88	32.2%	128.45	188.43	19.1%
Q3 2030	Model, 30% branch	1,107.32	356.49	32.2%	133.98	222.50	20.1%
Q4 2030	Model, 25% branch	1,058.89	340.98	32.2%	132.42	208.56	19.7%

Calendar-year model cross-check

Calendar year	Revenue	Gross profit	Opex	GAAP OP	OP margin
2027	2,385.83	650.22	345.01	305.21	12.8%
2028	2,968.31	847.91	401.09	446.82	15.1%
2029	3,505.99	1,075.14	458.75	616.39	17.6%
2030	4,093.78	1,318.01	521.45	796.56	19.5%

PENG: guidance changes economics; later persistence fades



Model chart: calendar-year reference and active revenue and GAAP operating profit. Fiscal guidance covers different quarters. Source: active PENG model, October 7, 2026.

5. Forward-four-quarter operating ceiling

Each ceiling uses the labeled quarter plus the next three quarters of active GAAP OP x20, divided by the labeled-quarter economic shares. Probabilities have already been applied to incremental operating economics and hedge credit; no second probability multiplier is applied. The final three operating quarters lack a complete forward window, so no ceiling is shown.

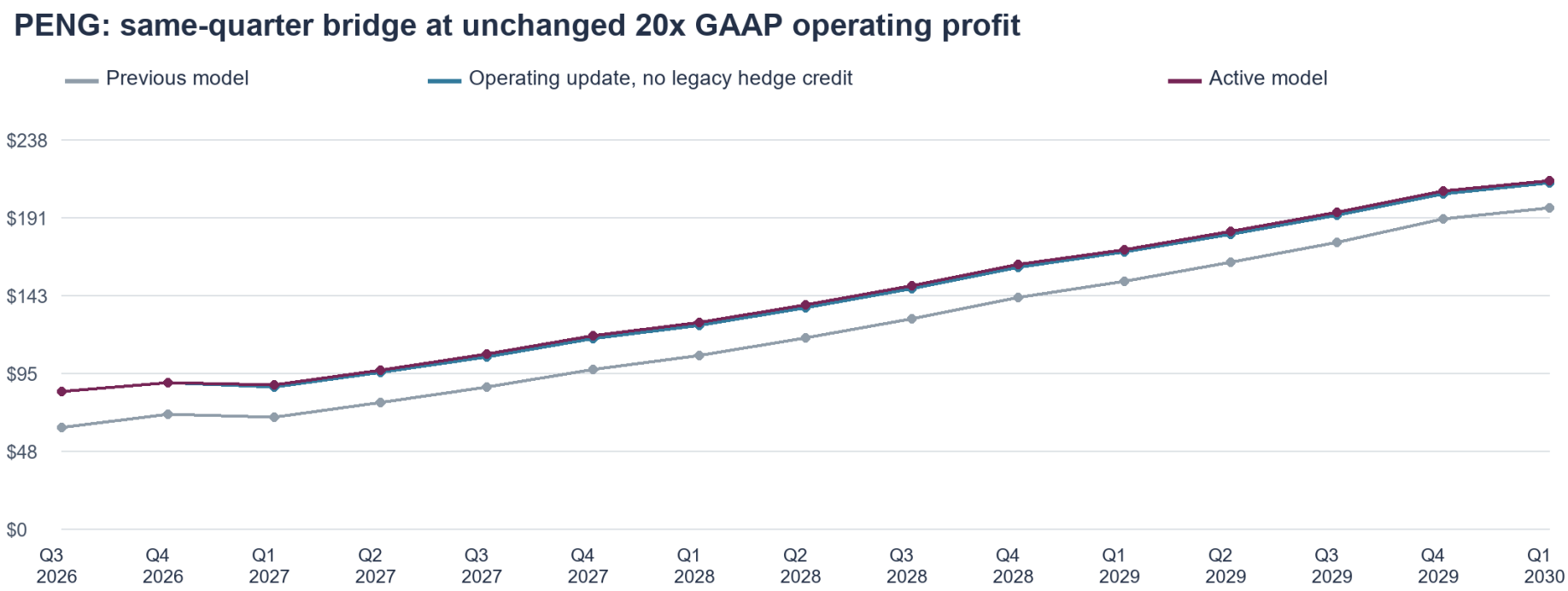
Quarter	Exact earnings window	F4Q GAAP OP	Multiple	Economic shares (m)	Ceiling (USD)
Q3 2026	Q3 2026 to Q2 2027	284.92	20x	68.04	83.75
Q4 2026	Q4 2026 to Q3 2027	303.52	20x	68.04	89.22
Q1 2027	Q1 2027 to Q4 2027	305.21	20x	69.48	87.86
Q2 2027	Q2 2027 to Q1 2028	337.45	20x	69.48	97.14
Q3 2027	Q3 2027 to Q2 2028	371.42	20x	69.48	106.91

Quarter	Exact earnings window	F4Q GAAP OP	Multiple	Economic shares (m)	Ceiling (USD)
Q4 2027	Q4 2027 to Q3 2028	410.55	20x	69.48	118.18
Q1 2028	Q1 2028 to Q4 2028	446.82	20x	70.81	126.20
Q2 2028	Q2 2028 to Q1 2029	485.49	20x	70.81	137.12
Q3 2028	Q3 2028 to Q2 2029	526.20	20x	70.81	148.62
Q4 2028	Q4 2028 to Q3 2029	573.04	20x	70.81	161.85
Q1 2029	Q1 2029 to Q4 2029	616.39	20x	72.19	170.76
Q2 2029	Q2 2029 to Q1 2030	656.29	20x	72.19	181.81
Q3 2029	Q3 2029 to Q2 2030	698.33	20x	72.19	193.46
Q4 2029	Q4 2029 to Q3 2030	746.79	20x	72.19	206.88
Q1 2030	Q1 2030 to Q4 2030	796.56	20x	74.83	212.89

Same-quarter change and dilution sensitivity

Q4 2027 bridge	USD ceiling	Meaning
Previous saved model	97.29	Same quarter/currency/basis
Operating update, prior denominator	116.30	No legacy hedge credit
Active model	118.18	90% legacy credit once
Active, 5% more shares	112.55	Mechanical sensitivity
Active, 10% more shares	107.44	Mechanical sensitivity

The active Q4 2027 earnings window contains $410.554m GAAP OP and 69.478673m economic shares. Same-quarter ceiling change is 21.47%. Changing the old artifact's Q1 2028 headline to the current Q4 2027 horizon is a presentation roll, excluded from this economic-change percentage.



Model chart: previous model, operating-only update and active case at identical quarterly horizons. Basis remains 20x GAAP OP. Hedge economics are counted once.

6. Industry relationships and read-throughs

MRVL, ALAB and RMBS can inform memory/interconnect architecture, while Dell provides a system-integration comparison. Memory producer pricing is an input-cost signal for Penguin. Do not transfer a producer's scarcity margin directly to a reseller, integrator or net-fee logistics model.

7. Risks and falsification tests

The following ledger makes the existing aggregate risk deduction explicit. It does not introduce another deduction or change the analytical score.

Risk class	Existing deduction
Hardware and service margin capture	-1
Memory-price and replacement-cost cycle	-1
Inventory, receivables and working capital	-1

Risk class	Existing deduction
Financing and diluted shares	-1
Product commercialization and execution	-1
Total	**-5**

Score and next evidence

Ceiling is probability-weighted forward-four-quarter GAAP OP x20 divided by the labeled-quarter economic-share proxy. Incremental economics are weighted once; the finished ceiling is not multiplied by probability again. Next evidence is the FY26 10-K, final exchange shares and surviving hedge terms, memory price-versus-volume growth, acceptance and collections, service attachments and paid MemoryAI orders. Shared learning: memory inflation can help module inventory margins temporarily while raising costs in the same company's AI hardware business.

Numbered monitoring tests

#	Checkpoint	Evidence supporting the operating case	Counter-test / failure signal
1	Revenue recognition	Reconcile gross logistics throughput to recognized net fees and principal sales.	Counting throughput as revenue overstates both market capture and reported scale.
2	MemoryAI adoption	Track paid deployments and workload-specific customer evidence.	Benchmarks without commercial conversion do not validate the software ramp.
3	Guidance and profit	Reconcile fiscal guidance, product GP and GAAP/adjusted OP.	Higher hardware sales without margin conversion does not establish software-like economics.
4	Cash and shares	Track inventory financing, collections and July financing consequences.	Weak cash conversion or added issuance changes per-share capture.

8. Sources, evidence gaps and model snapshot

The register distinguishes issuer facts, guidance and analyst assumptions. Historical source commentary remains attributed; integration does not independently verify every old claim. Detailed product capacity, subsegment profits and future pricing remain less certain than consolidated accounts.

- **PENG-1** | July 7, 2026 | Q3 FY26 release. Previous guidance.
- **PENG-2** | July 7, 2026 | Form 10-Q. Accounts, common and preferred terms.

- **PENG-3** | July 17, 2026 | Financing 8-K. 2031 notes, exchange shares and calls.
- **PENG-4** | July 7, 2026 | Issuer Q3 remarks and Q&A. Earlier management discussion.
- **PENG-5** | April 2, 2026 | Q2 public transcript. Historical procurement/pricing discussion; secondary transcription.
- **PENG-6** | FY2025 filing | Form 10-K. Annual figures and legacy note/call terms.
- **PENG-7** | September 6, 2026 saved artifact | User-approved model and sanctioned 20x GAAP OP valuation. Recoverable before snapshot retained.
- **PENG-8** | October 6, 2026 | Q4/FY26 issuer results. Actuals, guidance, cash flow and balance sheet.
- **PENG-9** | October 6, 2026 | Original issuer presentation, preserved in DOT-EARNINGS-20261006T1924 revision 1. Slides 13, 25, 27 and 32: customer mix caveat, gross logistics denominator, guidance and hedge evidence. Guidance and legacy-hedge panels are embedded here.
- **PENG-10** | October 6, 2026 | Issuer Q4 prepared remarks. Full prepared remarks reviewed; limited secondary Q&A is not a full primary transcript.
- **PENG-11** | July 13, 2026 | Issuer financing proposal. Intent to retain old calls; future unwind discretion preserved.
- **PENG-12** | October 6, 2026 | CFO appointment 8-K and exhibit preserved in the same immutable packet. Compensation terms, not an immediate grant-value expense.

Material unresolved items

- FY26 10-K and exact post-exchange common count; the 63m guidance is weighted average, not a complete capital reconciliation.
- Final legacy note/call settlement and survival. The 90% hedge recognition is explicit and bounded; cash elections and future counterparties can alter economic shares.
- Primary Q4 Q&A, exact backlog, memory price-versus-volume split, acceptance and quarterly FY27 recognition. Secondary Q&A excerpts remain qualitative.
- Inconsistent prior-year 60% AI mix. No historical numerical series is based on that claim.
- Product-level GM, paid MemoryAI volume and detailed working-capital funding forecast. Long-dated allocations are assumptions, not certified saleable capacity.
- CFO vesting, expense and actual grant shares; current overlap treatment avoids duplicating guided expense and the existing award proxy.

Model snapshot: October 7, 2026; 18 operating quarters Q3 2026-Q4 2030 and 15 ceilings through Q1 2030. Active Q4 2027 = USD 118.181332002; portfolio Base 73/100, analytical Base 72/100. Source packet DOT-EARNINGS-20261006T1924 revision 1; additional primary verification October 7. Recoverable source and model hashes are retained in the validation record.